Assignment #1

Requirements:

GPUs : 8 GPUs

usage: 8 hrs / day
22 days / month

Period: 4 years

if cloud, transfer of 5 TB / month

so, $8 \times 22 = 176$ hours / month

In 4 years:
$176 \times 48 = 8448$ hrs

on-prem TCO	cloud TCO
Each server = 4 GPUs so, servers needed = 2	instance = AWS p4d.24xlarge ↳ has 8 GPUs so 1 instance needed.
cost per server = \$ 65 800 For 2 servers = 65800×2 = \$ 131 600	cost = \$ 32.77 / hr cost of compute = 32.77×8448 = \$ 276 795
operating cost = \$ 5000 / yr / server For 4 yrs, 2 servers: $5000 \times 4 \times 2$ = \$40 000	Data transfer of 5 TB / month Aws rate = \$ 0.09 / GB 5 TB = 5000 GB
∴ Total on-prem TCO \$ 131 600 + \$40 000 = \$ 171,600	Monthly cost = 5000×0.09 = \$ 450 In 4 yrs = $12 \times 4 \times 450$ = \$ 21 600
∴ Total on-prem TCO is \$ 171 600.	Total cost = 21600 + 276 795 = \$ 298 395 ∴ Total cloud TCO is \$ 298 395.